

1 Z-Transform Properties

Property	Time-Domain Relation	Z-Domain Relation
Linearity	$ax_1[n] + bx_2[n]$	$aX_1(z) + bX_2(z)$
Time Shifting	$x[n - n_0]$	$z^{-n_0}X(z)$
Advance (negative shift)	$x[n + n_0]$	$z^{n_0}X(z)$
Scaling in amplitude	$ax[n]$	$aX(z)$
Multiplication by n	$nx[n]$	$-z \frac{dX(z)}{dz}$
Convolution in time	$x_1[n] * x_2[n]$	$X_1(z)X_2(z)$
Correlation	$\sum_n x[n]y^*[n - k]$	$X(z)Y^*\left(\frac{1}{z^*}\right)$
Time Reversal	$x[-n]$	$X\left(\frac{1}{z}\right)$
Exponential Scaling	$a^n x[n]$	$X\left(\frac{z}{a}\right)$
Differentiation in z	$nx[n]$	$-z \frac{dX(z)}{dz}$
Initial Value Theorem	$x[0]$	$\lim_{z \rightarrow \infty} X(z)$
Final Value Theorem	$\lim_{n \rightarrow \infty} x[n]$	$\lim_{z \rightarrow 1} (1 - z^{-1})X(z)$, if stable
Conjugation	$x^*[n]$	$X^*\left(\frac{1}{z^*}\right)$
Parseval's Relation	$\sum_n x[n]y^*[n]$	$\frac{1}{2\pi j} \oint X(z)Y^*\left(\frac{1}{z^*}\right) \frac{dz}{z}$
Decimation (by M)	$x[Mn]$	$\frac{1}{M} \sum_{k=0}^{M-1} X(z^{1/M} W_M^{-k})$
Upsampling (by M)	$\begin{cases} x[n/M], & n = mM \\ 0, & \text{otherwise} \end{cases}$	$X(z^M)$
Right-Sided Sequence	$x[n] = 0, n < 0$	ROC: $ z > \text{outermost pole}$
Left-Sided Sequence	$x[n] = 0, n > 0$	ROC: $ z < \text{innermost pole}$
Two-Sided Sequence	both sides nonzero	ROC: $r_1 < z < r_2$

2 DTFT

Property	Time Domain	Frequency Domain (DTFT)
Linearity	$ax_1[n] + bx_2[n]$	$aX_1(e^{j\omega}) + bX_2(e^{j\omega})$
Time Shift	$x[n - n_0]$	$e^{-j\omega n_0}X(e^{j\omega})$
Amplitude Scaling	$ax[n]$	$aX(e^{j\omega})$
Time Reversal	$x[-n]$	$X(e^{-j\omega})$
Frequency Shift	$e^{j\omega_0 n}x[n]$	$X(e^{j(\omega - \omega_0)})$
Convolution (Time)	$x_1[n] * x_2[n]$	$X_1(e^{j\omega})X_2(e^{j\omega})$
Multiplication (Time)	$x_1[n]x_2[n]$	$\frac{1}{2\pi} \int_{-\pi}^{\pi} X_1(e^{j\theta})X_2(e^{j(\omega - \theta)})d\theta$
Conjugation	$x^*[n]$	$X^*(e^{-j\omega})$
Parseval's Relation	$\sum_n x[n] ^2$	$\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) ^2 d\omega$
Differentiation in Frequency	$nx[n]$	$j \frac{d}{d\omega} X(e^{j\omega})$
Periodic Extension	$x[n]$ periodic with period N	$X(e^{j\omega}) = 2\pi \sum_k c_k \delta(\omega - 2\pi k/N)$
Real Sequence	$x[n]$ real	$X(e^{j\omega}) = X^*(e^{-j\omega})$ (conjugate symmet

3 DFT

Property	Time Domain	Frequency Domain (DFT)
Linearity	$ax_1[n] + bx_2[n]$	$aX_1[k] + bX_2[k]$
Circular Time Shift	$x[(n - n_0)_N]$	$W_N^{kn_0} X[k]$
Circular Frequency Shift	$W_N^{-k_0 n} x[n]$	$X[(k - k_0)_N]$
Circular Convolution	$x_1[n] \circledast x_2[n]$	$X_1[k] X_2[k]$
Pointwise Multiplication	$x_1[n] x_2[n]$	$\frac{1}{N} X_1[k] \circledast X_2[k]$
Conjugation	$x^*[n]$	$X^*[-k]_N$
Time Reversal	$x[-n]_N$	$X^*[k]_N$
Parseval's Theorem	$\sum_{n=0}^{N-1} x[n] ^2$	$\frac{1}{N} \sum_{k=0}^{N-1} X[k] ^2$
Real Sequence	$x[n]$ real	$X[k] = X^*[-k]_N$ (conjugate symmetry)
Even/Odd Symmetry	$x[n]$ even/odd	$X[k]$ real/imaginary
Time Differentiation	$nx[n]$	$j \frac{N}{2\pi} \frac{dX[k]}{dk}$ (approximate)